

LINKED LIST

GATE CSE 2026 SET-1

Que 1 :

Consider the following code snippet in C language that computes the number of nodes in a non-empty singly linked list pointed to by the pointer variable head.

```
struct node{  
  
    int elt;  
  
    struct node *next;  
  
};  
  
int getListSize (struct node *head)  
{  
  
    if( E1 ) return 1;  
  
    return E2 ;  
  
}
```

Which one of the following options gives the correct replacements for the expressions E1 and E2?

- 1) E1: head == NULL, E2: 1 + getListSize(head)
- 2) E1: head->next == NULL, E2: getListSize(head->next)
- 3) E1: head == NULL, E2: 1 + getListSize(head->next)
- 4) E1: head->next == NULL, E2: 1 + getListSize(head)

GATE CSE 2025 SET-2

Que 2 :

Consider an unordered list of N distinct integers.

What is the minimum number of element comparisons required to find an integer in the list that is NOT the largest in the list?

- 1) 1
- 2) $N-1$
- 3) N
- 4) $2N-1$

GATE CSE 2023

Que 3 :

Let SLLdel be a function that deletes a node in a singly-linked list given a pointer to the node and a pointer to the head of the list. Similarly, let DLLdel be another function that deletes a node in a doubly-linked list given a pointer to the node and a pointer to the head of the list.

Let n denote the number of nodes in each of the linked lists. Which one of the following choices is TRUE about the worst-case time complexity of SLLdel and DLLdel?

- A) SLLdel is $O(1)$ and DLLdel is $O(n)$
- B) Both SLLdel and DLLdel are $O(\log(n))$
- C) Both SLLdel and DLLdel are $O(1)$
- D) SLLdel is $O(n)$ and DLLdel is $O(1)$

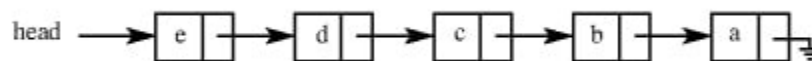
GATE CSE 2022

Que 4 :

Consider the problem of reversing a singly linked list. To take an example, given the linked list below,



the reversed linked list should look like



Which one of the following statements is TRUE about the time complexity of algorithms that solve the above problem in $O(1)$ space?

- A) The best algorithm for the problem takes $\theta(n)$ time in the worst case.
- B) The best algorithm for the problem takes $\theta(n \log n)$ time in the worst case.
- C) The best algorithm for the problem takes $\theta(n^2)$ time in the worst case.
- D) It is not possible to reverse a singly linked list in $O(1)$ space.

GATE CSE 2021 SET-2

Que 5 :

Consider the following ANSI C program:

```
#include <stdio.h>

#include <stdlib.h>

struct Node{
    int value;
    struct Node *next;};

int main() {
    struct Node *boxE, *head, *boxN; int index=0;
    boxE=head= (struct Node *) malloc(sizeof(struct Node));
    head -> value = index;
    for (index =1; index <= 3; index++){
        boxN = (struct Node *) malloc (sizeof(struct Node));
        boxE -> next = boxN;
        boxN -> value = index;
        boxE = boxN; }
    for (index=0; index <= 3; index++) {
        printf("Value at index %d is %d\n", index, head -> value);
        head = head -> next;
        printf("Value at index %d is %d\n", index+1, head -> value); }}
```

Which one of the following statements below is correct about the program?

- A) Upon execution, the program creates a linked-list of five nodes
- B) Upon execution, the program goes into an infinite loop
- C) It has a missing return statement which will be reported as an error by the compiler
- D) It dereferences an uninitialized pointer that may result in a run-time error

Que 6 :

What is the worst case time complexity of inserting n elements into an empty linked list, if the linked list needs to be maintained in sorted order?

- A) $O(n)$
- B) $O(n \log n)$
- C) $O(n^2)$
- D) $O(1)$

ISRO CSE 2018

Que 7 :

A doubly linked list is declared as:

```
struct Node {  
    int Value;  
    struct Node *Fwd;  
    struct Node *Bwd;  
};
```

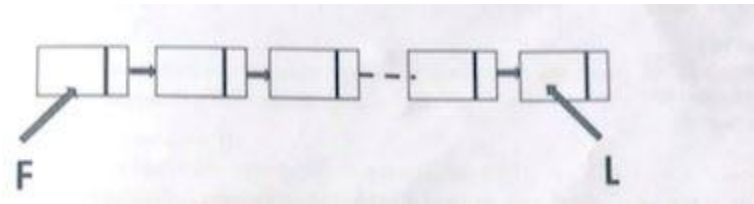
Where Fwd and Bwd represent forward and backward link to the adjacent elements of the list. Which of the following segment of code deletes the node pointed to by X from the doubly linked list, if it is assumed that X points to neither the first nor the last node of the list?

- A) $X \rightarrow Bwd \rightarrow Fwd = X \rightarrow Fwd; X \rightarrow Fwd \rightarrow Bwd = X \rightarrow Bwd;$
- B) $X \rightarrow Bwd.Fwd = X \rightarrow Fwd; X.Fwd \rightarrow Bwd = X \rightarrow Bwd;$
- C) $X.Bwd \rightarrow Fwd = X.Bwd; X \rightarrow Fwd.Bwd = X.Bwd$
- D) $X \rightarrow Bwd \rightarrow Fwd = X \rightarrow Bwd; X \rightarrow Fwd \rightarrow Bwd = X \rightarrow Fwd;$

ISRO CSE 2018

Que 8 :

Consider a singly linked list of the form where F is a pointer to the first element in the linked list and L is the pointer to the last element in the list. The time of which of the following operations depends on the length of the list?



- A) Delete the last element of the list
- B) Delete the first element of the list
- C) Add an element after the last element of the list
- D) Interchange the first two elements of the list

ISRO CSE 2017

Que 9 :

In a doubly linked list the number of pointers affected for an insertion operation will be

- A) 4
- B) 0
- C) 1
- D) Depends on the nodes of the doubly linked list

ISRO CSE 2017

Que 10 :

Given two statements

Insertion of an element should be done at the last node of the circular list

Deletion of an element should be done at the last node of the circular list

- A) Both are true
- B) Both are false
- C) First is false and second is true
- D) None of the above

GATE CSE 2017 SET-1

Que 11 :

Consider the C code fragment given below.

```
typedef struct node {  
    int data;  
    node* next ;  
} node;  
  
void join (node* m, node* n) {  
    node* p=n ;  
    while (p->next !=NULL){  
        p = p -> next ;  
    }  
    p-> next = m;  
}
```

Assuming that m and n point to valid NULL- terminated linked lists, invocation of join will

- A) append list m to the end of list n for all inputs.
- B) either cause a null pointer dereference or append list m to the end of list n.
- C) cause a null pointer dereference for all inputs.
- D) append list n to the end of list m for all inputs.

GATE CSE 2016 SET-2

Que 12 :

N items are stored in a sorted doubly linked list. For a delete operation, a pointer is provided to the record to be deleted. For a decrease-key operation, a pointer is provided to the record on which the operation is to be performed.

An algorithm performs the following operations on the list in this order: $\Theta(N)$ delete , $O(\log N)$ insert, $O(\log N)$ find, and $\Theta(N)$ decrease – key. What is the time complexity of all these operations put together?

- A) $O(\log^2 n)$
- B) $O(n)$

- C) $O(n^2)$
- D) $O(n^2 \log n)$

GATE CSE 2015 SET-2

Que 13 :

An unordered list contains n distinct elements. The number of comparisons to find an element in this list that is neither maximum nor minimum is

- A) $O(n \log n)$
- B) $O(n)$
- C) $O(\log n)$
- D) $O(1)$

GATE CSE 2015 SET-1

Que 14 :

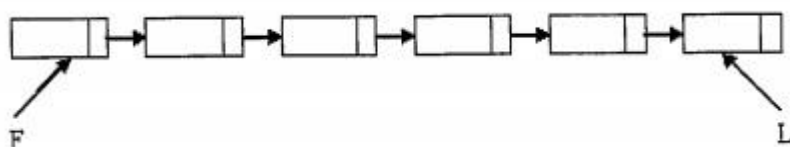
An algorithm performs $(\log N)^{1/2}$ find operations, N insert operations, $(\log N)^{1/2}$ delete operations, and $(\log N)^{1/2}$ decrease-key operations on a set of data items with keys drawn from a linearly ordered set. For a delete operation, a pointer is provided to the record that must be deleted. For the decrease-key operation, a pointer is provided to the record that has its key decreased. Which one of the following data structures is the most suited for the algorithm to use, if the goal is to achieve the best total asymptotic complexity considering all the operations?

- A) Unsorted array
- B) Min-Heap
- C) Sorted array
- D) Sorted doubly linked list

ISRO CSE 2014

Que 15 :

Consider a single linked list where F and L are pointers to the first and last elements respectively of the linked list. The time for performing which of the given operations depends on the length of the linked list?



- A) Delete the first element of the list
- B) Interchange the first two elements of the list
- C) Delete the last element of the list
- D) Add an element at the end of the list

ISRO CSE 2013

Que 16 :

The following steps in a linked list

```
p = getnode()
```

```
info(p) = 10
```

```
next (p) = list
```

```
list = p
```

result in which type of operation?

- A) Pop operation in a stack
- B) Removal of a node
- C) Inserting a node
- D) Modifying an existing node

GATE CSE 2010

Que 17 :

The following C function takes a simply-linked list as input argument. It modifies the list by moving the last element to the front of the list and returns the modified list. Some part of the code is left blank.

```
typedef struct node {  
    int value;  
    struct node *next;  
} Node;  
  
Node *move_to_front(Node *head) {  
    Node *p, *q;  
    if ((head == NULL) || (head->next == NULL)) return head;
```



```
q = NULL; p = head;
while (p->next !=NULL) {
    q=P;
    p=p->next;
}
```

```
return head;
}
```

Choose the correct alternative to replace the blank line.

- A) q = NULL; p→next = head; head = p;
- B) q→next = NULL; head = p; p→next = head;
- C) head = p; p→next = q; q→next = NULL;
- D) q→next = NULL; p→next = head; head = p;

ISRO CSE 2008

Que 18 :

The minimum number of fields with each node of doubly linked list is

- A) 1
- B) 2
- C) 3
- D) 4

ISRO CSE 2008

Que 19 :

Which of the following operations is performed more efficiently by doubly linked list than by linear linked list?

- A) Deleting a node whose location is given
- B) Searching an unsorted list for a given item
- C) Inserting a node after the node with a given location
- D) Traversing the list to process each node

ISRO CSE 2008

Que 20 :

The time required to search an element in a linked list of length n is

- A) $O(\log n)$
- B) $O(n)$
- C) $O(n^2)$
- D) $O(1)$

STACK

GATE CSE 2026 SET-2

Que 21 :

Consider a stack S and a queue Q . Both of them are initially empty and have the capacity to store ten elements each. The elements 1,2,3,4, and 5 arrive one by one, in that order. When an element arrives, it is assigned either to S (pushed on S) or to Q (enqueued to Q). Once all the five elements are stored, the output is generated in two steps. First, stack S is emptied by popping all elements. Then queue Q is emptied by dequeuing all elements. The output obtained by following this process is 4 3 1 2 5.

Given the output, the objective is to predict whether an element was assigned to S or Q .

Which of the following options is/are possible valid assignment(s) of the elements?

Note: In the options, the notation xS denotes that element x was assigned to S and yQ denotes that element y was assigned to Q .

- A) 1S,2Q,3S,4S,5Q
- B) 1Q,2Q,3S,4S,5Q
- C) 1Q,2Q,3Q,4S,5S
- D) 1S,2S,3S,4Q,5Q

GATE CSE 2025 SET-2

Que 22 :

Consider a stack data structure into which we can PUSH and POP records. Assume that each record pushed in the stack has a positive integer key and that all keys are distinct. We wish to augment the stack data structure with an $O(1)$ time MIN operation that

returns a pointer to the record with the smallest key present in the stack

- 1) without deleting the corresponding record, and
- 2) without increasing the complexities of the standard stack operations.

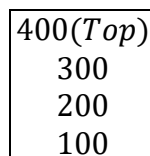
Which one or more of the following approach(es) can achieve it?

- A) Keep with every record in the stack, a pointer to the record with the smallest key below it.
- B) Keep a pointer to the record with the smallest key in the stack.
- C) Keep an auxiliary array in which the key values of the records in the stack are maintained in sorted order.
- D) Keep a Min-Heap in which the key values of the records in the stack are maintained.

GATE CSE 2024 SET-2

Que 23 :

Let S1 and S2 be two stacks. S1 has capacity of 4 elements. S2 has capacity of 2 elements. S1 already has 4 elements: 100, 200, 300, and 400, whereas S2 is empty, as shown below.



Stack S1



Stack S2

Only the following three operations are available:

PushToS2: Pop the top element from S1 and push it on S2.

PushToS1: Pop the top element from S2 and push it on S1.

GenerateOutput: Pop the top element from S1 and output it to the user.

Note that the pop operation is not allowed on an empty stack and the push operation is not allowed on a full stack.

Which of the following output sequences can be generated by using the above operations?

- A) 100, 200, 400, 300
- B) 200, 300, 400, 100
- C) 400, 200, 100, 300
- D) 300, 200, 400, 100

GATE CSE 2024 SET-1

Que 24 :

Consider the operator precedence and associativity rules for the integer arithmetic operators given in the table below.

<i>Operator</i>	<i>Precedence</i>	<i>Associativity</i>
+	<i>Highest</i>	<i>Left</i>
−	<i>High</i>	<i>Right</i>
*	<i>Medium</i>	<i>Right</i>
/	<i>Low</i>	<i>Right</i>

The value of the expression $3 + 1 + 5 * 2/7 + 2 - 4 - 7 - 6/2$ as per the above rules is

GATE CSE 2021 SET-1

Que 25 :

Consider the following sequence of operations on an empty stack.

push(54); push(52); pop(); push(55); push(62); s=pop();

Consider the following sequence of operations on an empty queue.

enqueue(21); enqueue(24); dequeue(); enqueue(28); enqueue(32); q=dequeue();

The value of s+q is _____.

ISRO CSE 2020

Que 26 :

A stack is implemented with an array of 'A[0 ... N - 1]' and a variable 'pos'. The push and pop operations are defined by the following code.

```

push (x)

A[pos] <- x

pos <- pos + 1

end push

pop()

pos <- pos + 1

return A[pos]

end pop

```

Which of the following will initialize an empty stack with capacity N for the above implementation?

- A) $\text{pos} \leftarrow -1$
- B) $\text{pos} \leftarrow 0$
- C) $\text{pos} \leftarrow 1$
- D) $\text{pos} \leftarrow N-1$

ISRO CSE 2020

Que 27 :

Convert the pre-fix expression to in-fix $-^* + ABC^* - DE + FG$

- A) $(A-B)*C+(D*E)-(F+G)$
- B) $(A+B)*C-(D-E)*(F-G)$
- C) $(A+B-C)*(D-E)*(F+G)$
- D) $((A+B)*C)-((D-E)*(F+G))$

ISRO CSE 2017

Que 28 :

Choose the equivalent prefix form of the following expression

$$(a + (b - c))^* ((d - e) / (f + g - h))$$

- A) $*+a-bc/-de-+fgh$
- B) $*+a-bc-/de-+fgh$
- C) $*+a-bc/-ed-+fgh$
- D) $*+ab-c/-de-+fgh$

ISRO CSE 2017

Que 29 :

The best data structure to check whether an arithmetic expression has balanced parenthesis is a:

- A) Queue
- B) Stack
- C) Tree
- D) List

ISRO CSE 2016

Que 30 :

The following postfix expression with single digit operands is evaluated using a stack:

$$8\ 2\ 3\ ^\wedge / 2\ 3\ * + 5\ 1\ * -$$

Note that $^\wedge$ is the exponentiation operator. The top two elements of the stack after the first $*$ is evaluated are

- A) 6,1
- B) 5,7
- C) 3,2
- D) 1,5

ISRO CSE 2015

Que 31 :

If the sequence of operations - push (1), push (2), pop, push (1), push (2), pop, pop, pop, push (2), pop are performed on a stack, the sequence of popped out values

- A) 2,2,1,1,2
- B) 2,2,1,2,2
- C) 2,1,2,2,1
- D) 2,1,2,2,2

GATE CSE 2015 SET-3

Que 32 :

The result evaluating the postfix expression $10\ 5 + 60\ 6 / * 8 -$ is

- A) 284
- B) 213
- C) 142
- D) 71

ISRO CSE 2014

Que 33 :

The five items: A, B, C, D, and E are pushed in a stack, one after other starting from A. The stack is popped four items and each element is inserted in a queue. The two elements are deleted from the queue and pushed back on the stack. Now one item is popped from the stack. The popped item is

- A) A
- B) B
- C) C
- D) D

GATE CSE 2014 SET-2

Que 34 :

Suppose a stack implementation supports an instruction REVERSE, which reverses the order of elements on the stack, in addition to the PUSH and POP instructions. Which one of the following statements is TRUE with respect to this modified stack?

- A) A queue cannot be implemented using this stack.
- B) A queue can be implemented where ENQUEUE takes a single instruction and DEQUEUE takes a sequence of two instructions.
- C) A queue can be implemented where ENQUEUE takes a sequence of three instructions and DEQUEUE takes a single instruction
- D) A queue can be implemented where both ENQUEUE and DEQUEUE take a single instruction each.

ISRO CSE 2009

Que 35 :

The five items: A, B, C, D, and E are pushed in a stack, one after other starting from A. The stack is popped four items and each element is inserted in a queue. The two elements are deleted from the queue and pushed back on the stack. Now one item is popped from the stack. The popped item is

- A) A
- B) B
- C) C
- D) D

Queue

GATE CSE 2023

Que 36 :

Consider a sequence a of elements $a_0 = 1, a_1 = 5, a_2 = 7, a_3 = 8, a_4 = 9, \text{ and } a_5 = 2$. The following operations are performed on a stack S and a queue Q , both of which are initially empty.

I: push the elements of a from a_0 to a_5 in that order into S .

II: enqueue the elements of a from a_0 to a_5 in that order into Q .

III: pop an element from S .

IV: dequeue an element from Q .

V: pop an element from S .

VI: dequeue an element from Q .

VII: dequeue an element from Q and push the same element into S .

VIII: Repeat operation VII three times.

IX: pop an element from S .

X: pop an element from S .

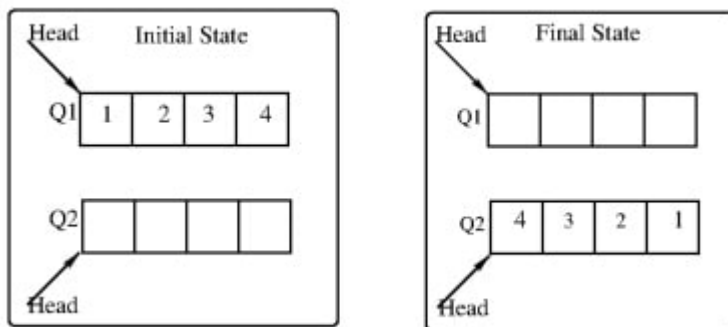
The top element of S after executing the above operations is ____.

GATE CSE 2022

Que 37 :

Consider the queues Q1 containing four elements and Q2 containing none (shown as the Initial State in the figure). The only operations allowed on these two queues are $Enqueue(Q, element)$ and $Dequeue(Q)$. The minimum number

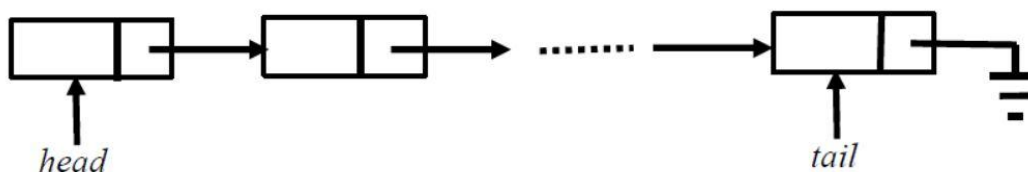
of *Enqueue* operations on Q1 required to place the elements of Q1 in Q2 in reverse order (shown as the Final State in the figure) without using any additional storage is



GATE CSE 2018

Que 38 :

A queue is implemented using a non-circular singly linked list. The queue has a head pointer and a tail pointer, as shown in the figure. Let n denote the number of nodes in the queue. Let enqueue be implemented by inserting a new node at the head, and dequeue be implemented by deletion of a node from the tail.



Which one of the following is the time complexity of the most time-efficient implementation of enqueue and dequeue, respectively, for this data structure?

- A) $O(1), O(1)$
- B) $O(1), O(n)$
- C) $O(n), O(1)$
- D) $O(n), O(n)$

ISRO CSE 2017

Que 39 :

Which of the following data structure is useful in traversing a given graph by breadth first search?

- A) Stack
- B) Queue
- C) List

D) None of the above

GATE CSE 2017 SET-2

Que 40 :

A circular queue has been implemented using a single linked list where each node consists of a value and a single pointer pointing to the next node. We maintain exactly two external pointers FRONT and REAR pointing to the front node and the rear node of the queue, respectively. Which of the following statements is/are CORRECT for such a circular queue, so that insertion and deletion operation can be performed in $O(1)$ time ?

- I. Next pointer of front node points to the rear node.
- II. Next pointer of rear node points to the front node.

- A) | only
- B) || only
- C) Both | and ||
- D) Neither | nor ||

GATE CSE 2016 SET-1

Que 41:

Let Q denote a queue containing sixteen numbers and S be an empty stack. $\text{Head}(Q)$ returns the element at the head of the queue Q without removing it from Q . Similarly $\text{Top}(S)$ returns the element at the top of S without removing it from S . Consider the algorithm given below.

```
while  $Q$  is not Empty do
  if  $S$  is Empty OR  $\text{Top}(S) \leq \text{Head}(Q)$  then
     $x := \text{Dequeue}(Q)$ ;
    Push( $S, x$ );
  else
     $x := \text{Pop}(S)$ ;
    Enqueue( $Q, x$ );
  end
end
```

The maximum possible number of iterations of the while loop in the algorithm is_____.

GATE CSE 2016 SET-1

Que 42 :

A queue is implemented using an array such that ENQUEUE and DEQUEUE operations are performed efficiently. Which one of the following statements is CORRECT (n refers to the number of items in the queue)?

- A) Both operations can be performed in $O(1)$ time
- B) At most one operation can be performed in $O(1)$ time but the worst case time for the other operation will be $\Omega(n)$
- C) The worst case time complexity for both operations will be $\Omega(n)$
- D) Worst case time complexity for both operations will be $\Omega(\log n)$